SE Lab-1

Smart Lab Equipment & Slot Reservation Portal

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SECTION: B

DATE: 19-08-2026

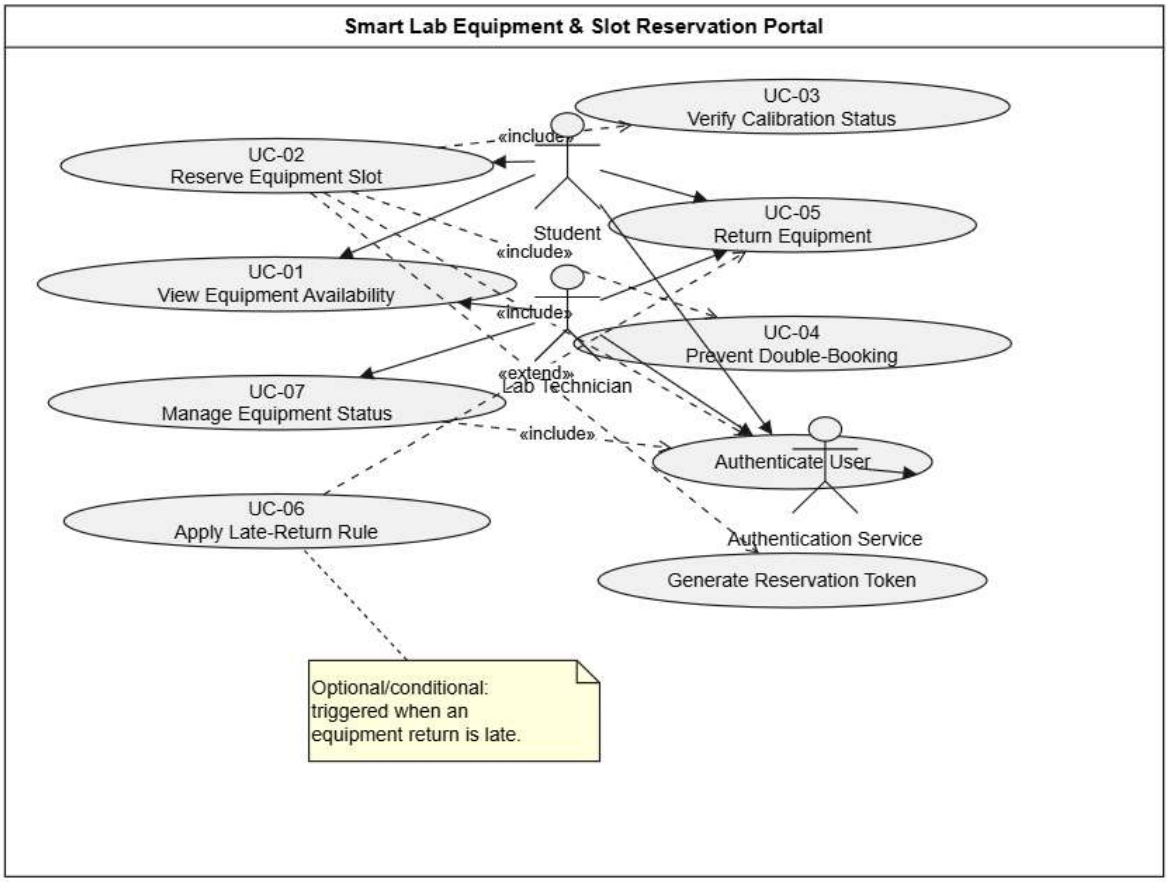
Deliverable 1 — Complete Requirements Table

5 Functional Requirements (FR-001 to FR-005) and 2 Non-Functional Requirements (NFR-001 to NFR-002), with all required fields.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow authenticated students to view real-time availability and reserve a maximum 2-hour slot for any calibrated equipment up to 7 days in advance.	High	Pass: A valid student can reserve an available calibrated item for ≤2 hours and ≤7 days ahead, and a reservation token is generated. Fail: A reservation exceeds 2 hours/7 days or double-booking occurs.	Core reservation requirement explicitly provided in the problem statement.
FR-002	Functional	The system shall verify the selected equipment's calibration status before confirming a reservation.	High	Pass: A reservation for calibrated equipment proceeds to confirmation; Fail: an uncalibrated item is blocked and the student is informed.	Prevents use of equipment whose calibration status does not meet the scenario constraint.
FR-003	Functional	The system shall prevent double-booking by locking a selected equipment slot before the reservation is confirmed.	High	Pass: Two concurrent requests for the same equipment and overlapping slot result in only one confirmed reservation. Fail: more than one reservation is confirmed for the same slot.	Required to prevent double-booking of high-value instruments.
FR-004	Functional	The system shall record equipment returns and apply the defined late-return disciplinary rules when an item is returned after its reservation end time.	Medium	Pass: An on-time return is recorded without a late penalty; a late return is recorded and the applicable disciplinary rule is triggered. Fail: late status is not detected or the rule is not applied.	The problem statement explicitly requires enforcement of late-return disciplinary rules.
FR-005	Functional	The system shall allow the Lab Technician to view reservations and update equipment availability and calibration status.	Medium	Pass: An authenticated Lab Technician can view reservation records and update equipment availability/calibration status, with the new status reflected in subsequent availability checks. Fail: the update is not saved/reflected.	Supports technician involvement in equipment and reservation operations.
NFR-001	Nonfunctional	The system shall ensure that concurrent slot reservation	High	Pass: Benchmarking under simulated peak concurrent load	Provided in the problem statement as

		requests are processed within 200 ms with strict database-level lock isolation to prevent race conditions.		records reservation processing at ≤200 ms and no race-condition double-bookings. Fail: latency exceeds 200 ms or isolation fails.	the performance and security requirement.
NFR-002	Nonfunctional	The system shall protect reservation and equipment-status data so that only authenticated users can perform actions permitted for their role.	High	Pass: Unauthenticated access is rejected and role-restricted actions are denied; authenticated Student and Lab Technician actions succeed only within their permitted functions. Fail: unauthorized action succeeds.	Security is central to authenticated reservation and technician operations.

Deliverable 2 — UML Use-Case Diagram



Deliverable 3 — Use-Case Flow

Preconditions

- The Student is authenticated.
- The requested equipment exists in the portal and its availability can be checked.
- The requested reservation is within the allowed advance window of 7 days.

Postconditions

- A valid reservation is recorded for the selected equipment and time slot.
- The reserved slot is locked so another reservation cannot double-book it.
- A reservation token is generated and presented to the Student.

Main Success Scenario

- Student selects “View Equipment Availability.”
- System displays real-time availability of lab equipment and slots.
- Student selects a calibrated equipment item and a desired time slot.
- System verifies that the equipment is calibrated and that the requested duration does not exceed 2 hours.
- System checks that the requested reservation is no more than 7 days in advance.
- System attempts to lock the selected equipment slot using database-level isolation.
- System confirms the reservation and generates a reservation token.
- System displays the reservation confirmation and token to the Student.

Alternate Flow

- A1. If the selected equipment is not calibrated, the System rejects the reservation and informs the Student to select calibrated equipment.
- A2. If the requested slot has already been locked/reserved, the System rejects the request and displays the current availability so the Student can choose another slot.
- A3. If the requested duration exceeds 2 hours or the requested date is more than 7 days ahead, the System rejects the request and asks the Student to provide a valid slot.